
Artificial Intelligence

Assignment 4

Total Points: 5

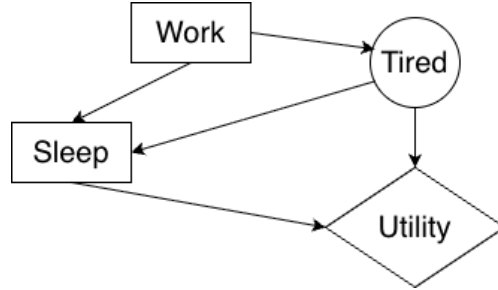
Uncertainty & Neural Nets

Due: 17:00, 8 May, 2026

Submission PDF to: <https://f.wps.cn/g/s0jhAbrX/>

1. Decision Networks (1.5 pts)

Give the optimal policy that achieves the highest expected utility of the following decision network:

**CPTs:**

Work	$P(\text{tired})$	Tired	Sleep	Utility
work	0.8	tired	sleep	30
rest	0.3	tired	no sleep	-10
		not tired	sleep	50
		not tired	no sleep	70

2. Markov Decision Process (2 pts)

A small robot must manage its energy level while working. At any time, the robot is in one of two **states**: High energy (H) and Low energy (L). In each state, the robot can choose one of two **actions**: Work (W) or Rest (R). The environment is stochastic. The **transition probabilities** $P(s' | s, a)$ are given below.

s	a	$P(s' = H s, a)$	$P(s' = L s, a)$
H	W	0.3	0.7
H	R	0.9	0.1
L	W	0.6	0.4
L	R	0.4	0.6

The immediate **reward function** $R(s, a)$ is provided by the table below.

s	a	Reward
H	W	+5
H	R	+1
L	W	+2
L	R	+3

Let the **discount factor** be $\gamma = 0.9$. Perform value iteration algorithm for **one iteration** starting from $V_0(H) = 0$ and $V_0(L) = 0$ and give the updated values $V_1(H), V_1(L)$ for the two states.

3. Neural Networks (1.5 pts) Consider a neural network to prediction sleep quality:

- Input layer: x_1, x_2 (Let x_1 be the average daily screen time, x_2 be daily physical activity)
- Hidden layer: 2 neurons with ReLU activation
- Output layer: 1 neuron with sigmoid activation, output 1 to denote good sleep quality, and 0 to denote bad sleep quality.

Weights of the hidden layer and the output layer (given):

$$W^{(h)} = \begin{bmatrix} 0.3 & -0.5 \\ 0.8 & 0.2 \end{bmatrix}, \quad b^{(h)} = \begin{bmatrix} 0.1 \\ -0.2 \end{bmatrix}, \quad W^{(out)} = [0.6 \quad -0.4], \quad b^{(out)} = 0.3$$

Consider the input sample: $x = [x_1, x_2]^T = [1.0, 0.5]^T$ with true label $y = 1$:

1. Compute hidden layer activations: $z = \text{ReLU}(W^{(h)}x + b^{(h)})$.
2. Compute output: $\hat{y}$.
3. Compute binary cross-entropy loss: $L = -[y \log \hat{y} + (1 - y) \log(1 - \hat{y})]$.